

ECG Anomaly Detection using Latent Representations and Reconstruction Error of a VAE

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Introduction

Manual analysis of ECG (electrocardiographic) signals is a costly, slow process that is highly dependent on specialists, which hinders its scalability in high-demand clinical settings. Given the growing need for diagnoses, we seek to develop automatic tools capable of reliably detecting anomalies.

Objectives

- Learn latent representations of normal ECG signals using a 1D VAE.
- Detect anomalies from the VAE reconstruction error.
- Classify anomalous signals using an XGBoost classifier over the latent vectors and the reconstruction score.
- Evaluate the pipeline's performance on the public PTB-XL and Chapman-Shaoxing databases.

Data & FMM

Datasets used

- PTB-XL ECG Dataset*: 21,837 12-lead records, 500 Hz.
- Chapman-Shaoxing ECG Dataset*: 10,646 multi-class arrhythmia records.

Morphological features (FMM) and preprocessing

21 coefficients were extracted that characterize the P, Q, R, S and T waves through the Frequency Modulated Möbius model, describing their amplitudes, phases, directionality and frequencies.

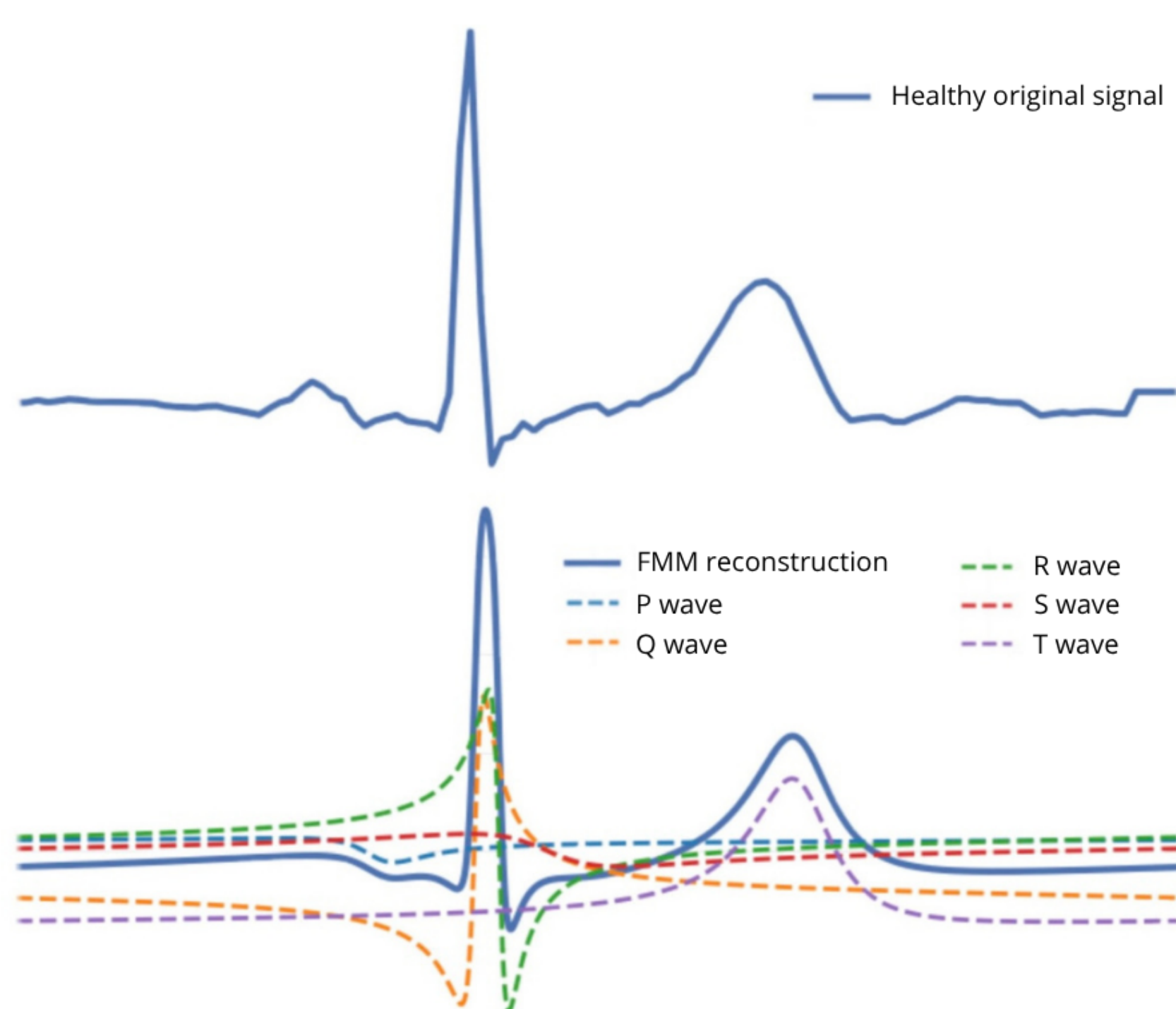


Figure 1: ECG signal and its reconstruction using the FMM model.

The ECG windows were resized to a fixed length of 2048 samples and each one was Z-score normalized using the statistics computed over the training set.

References

- [1] Verardo, Boman, Bruchfeld, Chiesa, Koch, Maguire Jr., and Kostic. *FMM-Head: Enhancing Autoencoder-based ECG anomaly detection with prior knowledge*, 2023.
- [2] Jang, Kim, Lim, and Yoon. *Unsupervised feature learning for electrocardiogram data using the convolutional variational autoencoder*, 2021.

Variational Autoencoder

The VAE compresses each ECG window together with its FMM coefficients into a latent space and then reconstructs it, so that the reconstruction error (ELBO) serves as an anomaly indicator.

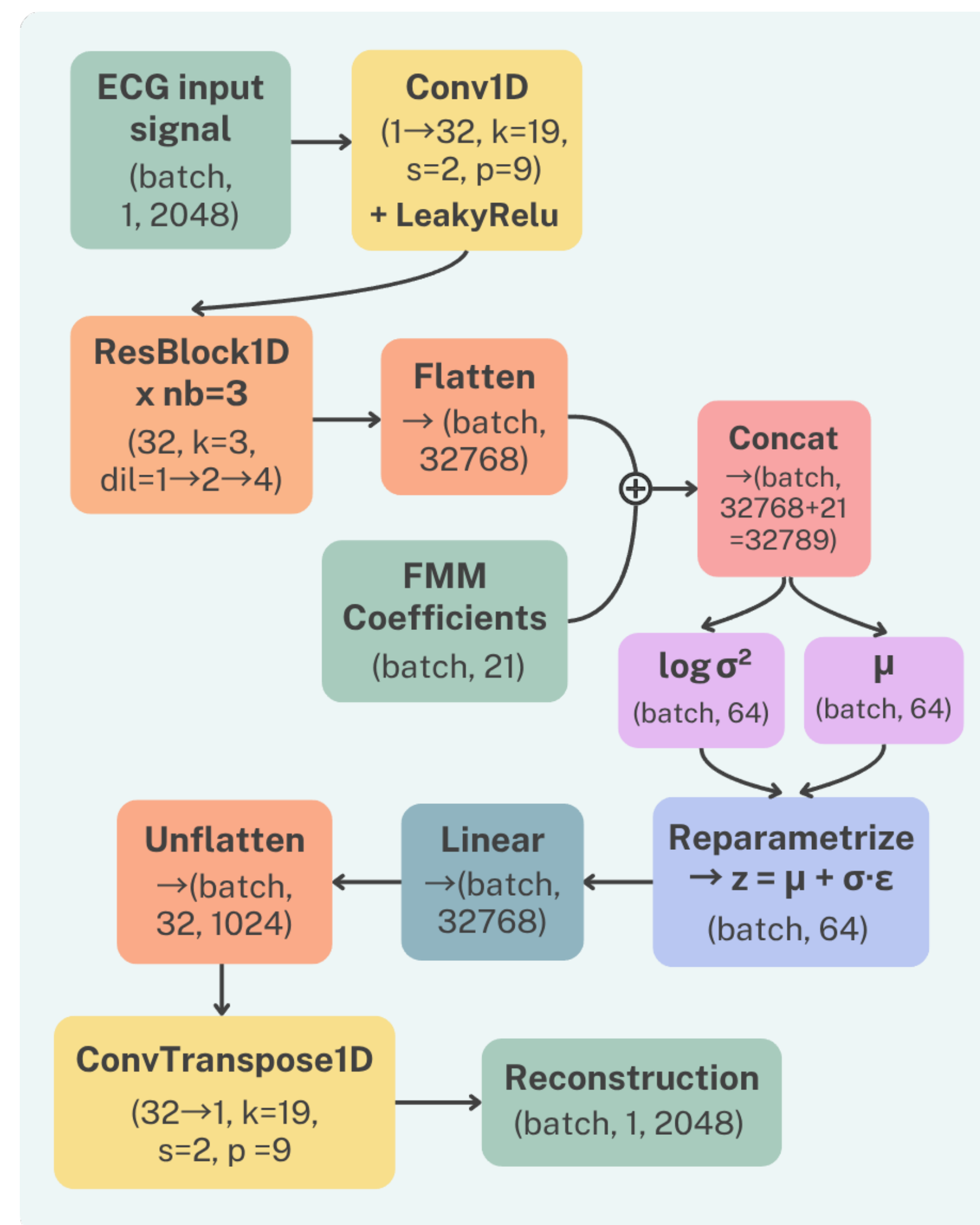


Figure 2: VAE architecture

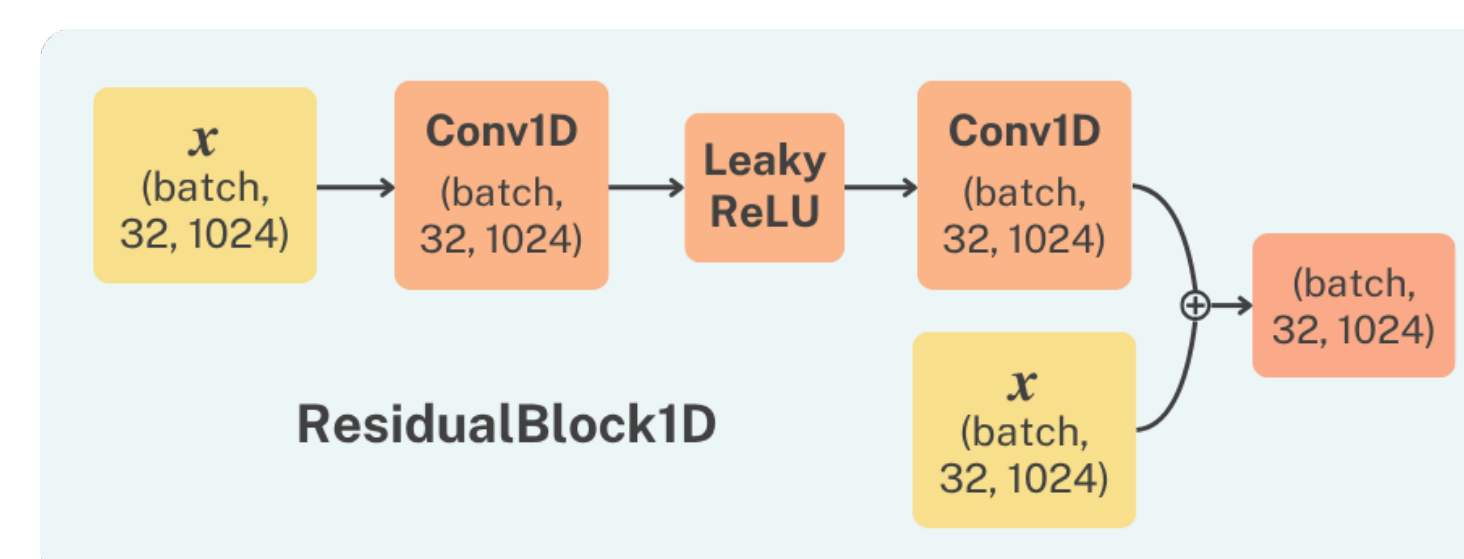


Figure 3: 1D Resize Block architecture

1. **Encoder:** takes as input the ECG signal (2048 samples) and the FMM coefficient vector. It extracts features and projects them into two latent parameter vectors, μ and $\log \sigma^2$.
2. **Reparameterization:** from μ and $\log \sigma^2$ we sample $z = \mu + \sigma \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$.
3. **Decoder:** takes z and generates the reconstruction $\hat{x}$ of the ECG window, trying to approximate the original signal.

Anomaly Score

The anomaly score is the ELBO, combining reconstruction error and latent regularization:

$$\text{ELBO} = \underbrace{-\text{MSE}(\hat{x}, x)}_{\text{reconstruction}} - \underbrace{\beta \text{KL}(q(z | x) \parallel \mathcal{N}(0, I))}_{\text{regularization}}.$$

XGBoost

For each beat we build $c = [\mu, e] \in \mathbb{R}^{65}$ from the latent mean μ and the reconstruction error e . An XGBoost classifier maps it to the final prediction (*Normal* vs. *Anomalous*).

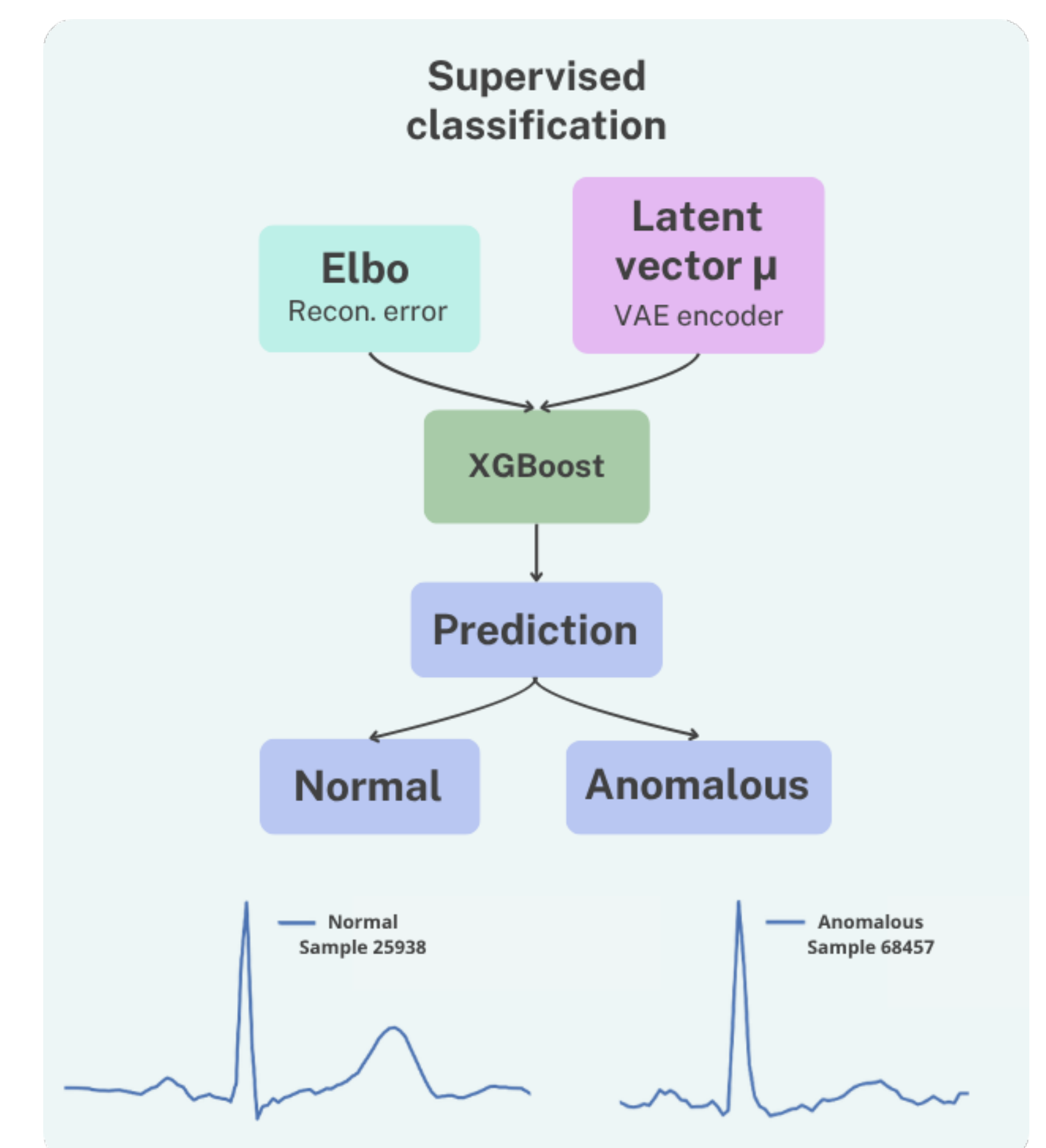
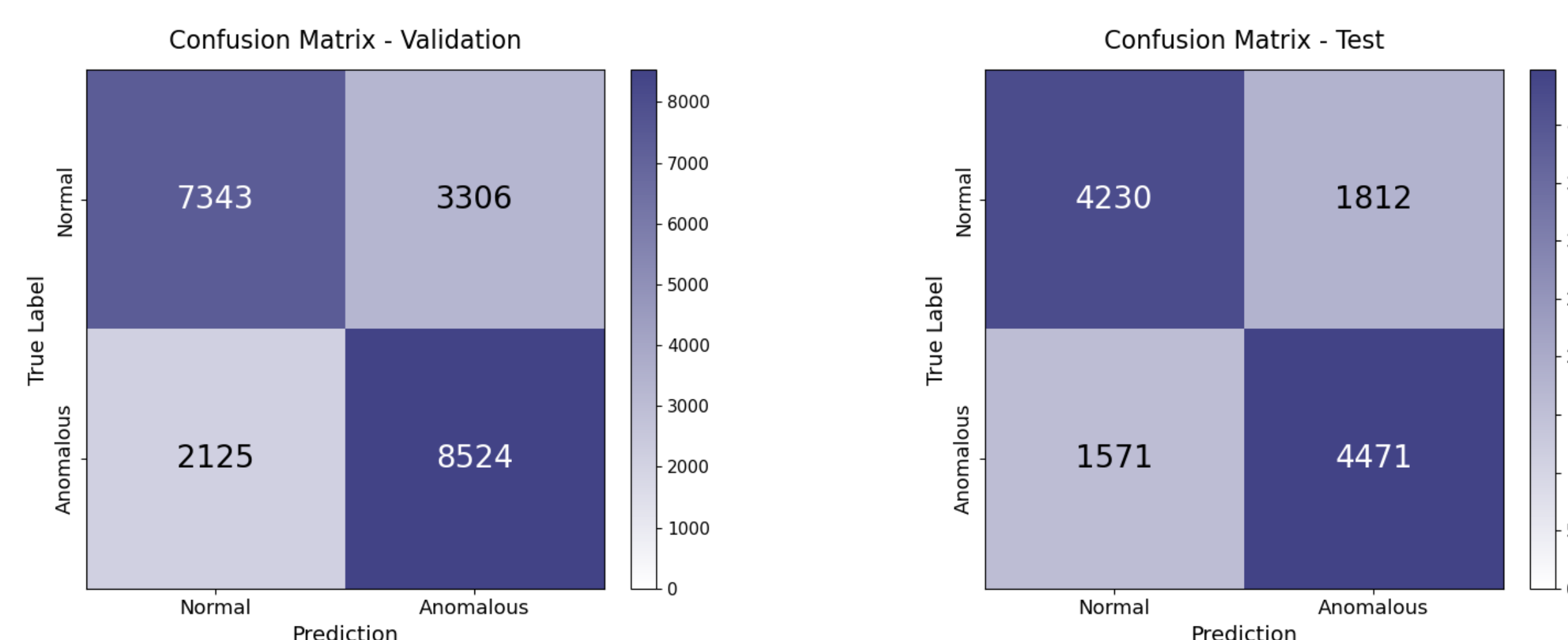


Figure 4: XGBoost classifier

Conclusions & Future Work

- The VAE + XGBoost pipeline recovers 74% of the anomalous beats on unseen patients.
- FMM coefficients add useful morphological prior knowledge.
- Splitting by recording instead of by beat is essential: it lowers validation AUC from 0.91 to 0.85.
- In the future, attention mechanisms or synthetic data could be incorporated.

Results



Confusion matrices: validation (left) and test (right)

All splits are grouped by recording, so no patient appears in more than one split. The threshold is picked on validation to keep recall above 0.80.

Metric	Validation	Test
AUC-ROC	0.8503	0.8058
Precision	0.7399	0.7116
Recall	0.8001	0.7400
F1-score	0.7688	0.7255
Accuracy	0.7594	0.7200